OHM'S LAW

- Was discovered by German scientist Georg Ohm in 1826
- He discovered that if you **double** the voltage across a conductor, **twice** the current will flow through it
- So, if you **triple** the voltage, **three** times the current will flow through it.
- Because he found this out it was called **OHMS LAW**
- **Ohms law** is;

The current flowing through a conductor is proportional to the voltage difference between its ends.

or

Provided the temperature remains constant, when a voltage is applied to a given resistance, the greater the voltage, the greater the current that flows. ***Note:*** *If the wire gets hot, the resistance changes (it increases)!*

- We can show this in a mathematical formula

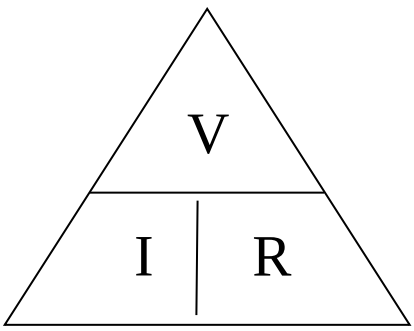
$$\frac{\text{Voltage}}{\text{Current}} = \text{Resistance}$$

Or by using

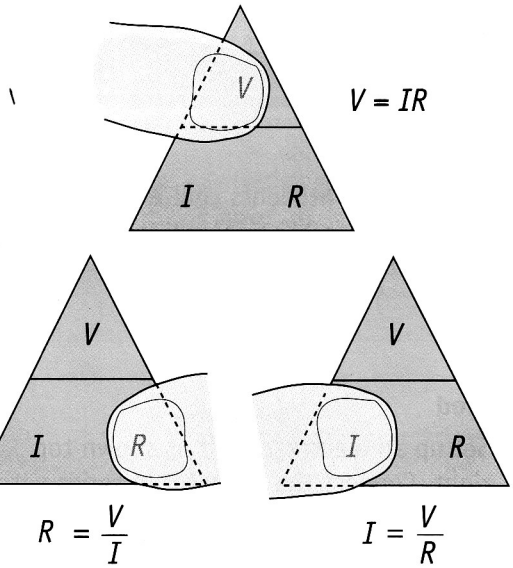
$$\frac{V}{I} = R$$

- **V = Voltage, measured in volts** (v)
- **I = current, measured amps** (A)
- **R = Resistance, measured in ohms (Ω)**

- So how can we work out current if we know the voltage and resistance
- We can use our equation triangle



A simple way to rearrange equations is to write the three variables in a triangle as shown. Cover the symbol that stands for what you want to find, and the other two symbols tell you how to calculate it.



- Let's try an example question…
  - If a 1.5 volt cell is connected to a lamp with a resistance of 10 ohms, what current will flow through the lamp?

What information do we have?	V = 1.5 v R = 10 Ω
What do we have to find out?	I = ?
What is the Ohms law equation	I = V / R
Substitute or information into equation	I = 1.5 / 10
Calculate the value of I	I = 0.15 A
Clearly state the answer	The current flowing in the lamp was 0.15 A

Your Turn

1. A current of 4.5 A flows through a resistor of 100 Ω. Calculate the voltage applied to the resistor.

Write the data down on the left hand side including units

$V = ?$
$I = 4.5\ A$
$R = 100\ \Omega$

$V = IR$
$= (4.5)(100)$
$=$

Set your working out clearly.

- Equation
- Substituted numbers
- Answer with units

2. If a 1.5 V cell is connected to a globe with a resistance of 10 Ω, what current will flow through it?

$V =$
$I =$
$R =$

$$I = \frac{V}{R}$$

3. A room heater working on 240 V uses 5.0 A of current when it is set on its highest setting. What is the resistance of the heater coil inside the machine?